

Hiep Tan To

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Bachelor's and Master's in Computer Science student at Ho Chi Minh University of Science, specializing in Software Engineering, Backend Development, and System Integration. Strong foundation in OOP, algorithms, and database systems. Experienced in building scalable pipelines, system-level applications, and contributing to open-source software.

Education

Master of Computer Science	In Progress
University of Science - VNUHCM	2025
Bachelor of Technology with Honors	9.41/10.0
Computer Science and Engineering, University of Science - VNUHCM	2021-2025
Thesis	10.0/10.0
Generative AI-Based Virtual KOL System	2025
Language Skills	IELTS 6.0, VSTEP B2

Internships and Experience

Software Engineer

Endava

May 2026 – Present

- Architect and scale a high-availability, high-capacity cluster platform supporting mission-critical global operations, ensuring system resilience and seamless service delivery across multiple regions.
- Collaborate with cross-functional teams to translate complex requirements into robust Linux-based system services, utilizing languages such as C/C++, Go, or Python to optimize low-level performance.
- Lead deep-dive troubleshooting and root cause analysis (RCA) for intricate system bugs reported by integration teams and global customers, reducing downtime and improving overall software stability.
- Drive continuous product evolution by refactoring legacy components and implementing modern architectural patterns, significantly enhancing system throughput and resource efficiency on Linux environments.
- Research and integrate emerging technologies into the cluster ecosystem to maintain a competitive edge and meet the increasing demands of high-traffic workloads.

Junior AI Engineer

First AI

Aug. 2025 - May. 2026

- Developed scalable, real-time recommendation systems for AI-driven search and social platforms, significantly improving content relevance and user engagement.
- Trained and fine-tuned machine learning and LLM-based models on large-scale user behavior and contextual datasets to improve ranking accuracy.
- Implemented retrieval-augmented generation (RAG) pipelines to enhance recommendation accuracy with domain-specific external knowledge bases.
- Integrated external LLM APIs to support semantic search and personalized recommendations.
- Optimized model inference for real-time recommendation services in production with minimal latency.
- Evaluated recommendation and model performance using metrics such as Precision@K, Recall@K, NDCG, and F1-score.

Integration Team, Treasure Data

Contractor Software Engineer

Apr 2024 – Aug 2025

- Engineered and optimized high-performance data pipelines and custom connectors using Embulk and REST APIs, ensuring seamless ETL workflows and high data integrity for large-scale enterprise datasets.
- Developed an intelligent AI document summarization agent by leveraging LLMs, implementing efficient prompt engineering and caching strategies that improved summarization throughput by 30%.
- Collaborated closely with cross-functional teams to architect scalable, reliable, and fault-tolerant data integration solutions, focusing on reducing latency and improving system modularity.
- Enhanced and modernized CLI manual testing tools and internal utilities, streamlining the QA lifecycle and enabling more rigorous validation of complex integration scenarios.

Undergraduate Research

Software Engineering Laboratory, University of Science

Sept. 2023 – Present

- Researched Generative AI techniques such as Stable Diffusion, GANs, and RAG.
- Contributed to successful publications, enhancing understanding of adversarial robustness and model defense strategies for NLP systems.
- Conducted research on physics-aware video generation, focusing on modeling physical phenomena such as melting and spring-mass systems.
- Benchmarked and optimized OCR models, including Surya and olmOCR, to improve accuracy for Vietnamese handwriting recognition tasks.

Technical Skills

- **Machine Learning / Deep Learning:** PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn, OpenCV
- **Generative AI:** Stable Diffusion, GANs, Retrieval-Augmented Generation (RAG), LangChain
- **Data Engineering / MLOps:** Embulk, Airflow, SQL, MongoDB, AWS (S3, EC2, Lambda)
- **Programming:** Python, C++, Java
- **Other Tools:** Streamlit, Docker, Git, Overleaf
- **Soft Skills:** Strong presentation, collaboration, and problem-solving

Scholastic Achievements

- **Graduated in the Top 5** of the cohort with **High Distinction** (GPA: 9.41/10.0, Perfect Thesis Score), earning **direct admission** to the Master's program in Computer Science at the University of Science.
- Awarded the prestigious **Certificate of Merit by the President of Vietnam National University** for exceptional academic and scientific research excellence.
- Awarded **Outstanding Student in Scientific Research Achievement** (2023).
- Honored in the **Top 100 Outstanding Students of Vietnam National University** (2023).
- Ranked **1st Best Student of Regular Program, Faculty of IT** (2021).
- Selected among **Top 5 Students** of the Faculty of IT (2021).
- Won **Third Prize** – Provincial Competition for Excellent Students in Informatics (2020).

Key Academic Projects

Embulk Encoder Encryption Plugin (Java, AES, Embulk Framework)

Role: *Creator & Developer*

2025

- Developed a comprehensive Embulk encoder plugin enabling seamless AES encryption of sensitive data before ETL pipeline output.
- Implemented configurable encryption algorithms (AES/CBC/PKCS5Padding) with flexible key and IV management strategies for enhanced security compliance.
- Engineered support for both text and binary data encryption with streaming and batch encoding modes for scalability.
- Designed seamless integration with existing Embulk output plugins, allowing encryption to be applied transparently within ETL workflows.
- Implemented custom key management and encoding options to accommodate diverse security requirements across different data export scenarios.
- GitHub: [embulk-encoder-encryption](#)

Embulk Input Marketo Plugin (Ruby, Embulk, Marketo REST API)

Role: *Contributor — Open-source Developer*

2024

- Contributed to an open-source Embulk plugin enabling data ingestion from Marketo using REST API.
- Improved support for bulk extraction of Leads, Activities, Campaigns, and Program Members with incremental ingestion.
- Collaborated on enhancing configuration parameters (batch size, polling interval, incremental filters) for scalability and reliability.
- GitHub: [embulk-input-marketo](#)

Remote Control PC (Java, Socket Programming)

Team size: 6 — Role: Developer, Project Manager

May 2022

- Built an application to remotely control a PC (process listing, app listing, screenshots, shutdown/restart).
- Applied socket programming and Java toolkits in a team project for the Computer Networking course.
- GitHub: [Remote_Control](#)

Chess Game (C/C++, SFML Library)

Team size: 5 — Role: Developer, Project Manager

Aug. 2022

- Developed a two-player chess game application for the Object-Oriented Programming course.
- Implemented game logic and user interface with SFML.
- GitHub: [Chess_OOP](#)

Additional Certificates

Completed FPT AI Compass

Completed AWS Academy Cloud Foundations

Completed Intro to Machine Learning (Kaggle)

Completed Unsupervised Learning, Recommenders, Reinforcement Learning (Coursera)

Completed Advanced Learning Algorithms (Coursera)

Completed Machine Learning Explainability (Kaggle)

Completed Machine Learning Specialization (Coursera)

Completed Ultimate AWS Certified AI Practitioner AIF-C01

Completed AWS Certified Developer Associate

Publication

GenKOL: Modular Generative AI Framework For Scalable Virtual KOL Generation

Team Size: 2

Jan. 2026

- Architected a scalable Generative AI framework built upon **Stable Diffusion** and custom-trained **LoRA** weights to automate the generation of diverse, high-fidelity Virtual KOLs fully.
- Engineered an ID-consistency preservation module utilizing advanced attention-injection and face-embedding techniques, ensuring stable visual identities across complex digital environments and dynamic poses.
- Optimized the end-to-end inference pipeline for large-scale deployment, significantly reducing character generation latency while maintaining photorealistic rendering quality.
- Peer-reviewed and accepted at **FJCAI 2026 - Can Tho** (The 3rd Future of Journalism and AI Conference).

Streamlining Virtual KOL Generation Through Modular Generative AI Architecture

Team Size: 2

Aug. 2025

- Designed a novel, multi-stage generative architecture integrating **ControlNet** and **IP-Adapter** to strictly control spatial structures and streamline the synthesis of highly realistic virtual human avatars.
- Implemented a robust, plug-and-play modular system supporting dense spatial conditioning (e.g., skeletal pose, depth, facial landmarks) to seamlessly switch between distinct visual styles and precise motion templates.
- Conducted comprehensive ablation studies demonstrating the superiority of the proposed modular architecture in achieving spatial accuracy and stylistic alignment over existing baselines.
- Presented research findings at **ACM Multimedia 2025** (Rank A*), showcasing significant computational efficiency advancements for AI-driven creative industries.

Budget-aware Road Semantic Segmentation in Unseen Foggy Scenes

Team Size: 3

Aug. 2023

- Proposed a highly efficient, budget-aware semantic segmentation model tailored for low-resource edge computing environments in autonomous driving systems.
- Leveraged Unsupervised **Domain Adaptation** alongside a Teacher-Student **Knowledge Distillation** framework to dynamically enhance model robustness against entirely unseen, heavily foggy weather conditions.
- Conducted extensive experiments on standard benchmark datasets (e.g., Foggy Cityscapes), achieving highly competitive Mean Intersection over Union (**mIoU**) scores without requiring computationally expensive retraining from scratch.
- Published and presented at the **RIVF 2023 International Conference**, demonstrating the model's capability to maintain real-time inference speed under severe weather degradation.
- Paper link: [IEEE Xplore](#)